

Nondimensionalization of CAR–T model

Copy-paste-ready L^AT_EX file containing the derivation and Python code.

Dimensional model (clean / explicit)

The dimensional model (as in the notebook) is

$$\frac{dT}{dt} = r T \left(1 - \frac{T}{K}\right) - \kappa \frac{C}{C + C_{50}} T, \quad (1)$$

$$\frac{dC}{dt} = p C \cdot \frac{T^{h_1}}{T^{h_1} + T_{50,1}^{h_1}} - \ell C - \delta_C C - q C \cdot \frac{T/C}{T/C + s}, \quad (2)$$

$$\frac{dM}{dt} = \ell C - b M \frac{T^{h_2}}{T^{h_2} + T_{50,2}^{h_2}} - \delta_M M. \quad (3)$$

Choice of characteristic scales

We choose the characteristic scales as discussed in the meeting and notes:

$$\text{Tumor scale: } K, \quad \text{Effector scale: } C_{50}, \quad \text{Memory scale: } C_{50}, \quad \text{Time scale: } \tau = \frac{1}{r}.$$

Define the dimensionless time and variables

$$s = rt, \quad x(s) = \frac{T(t)}{K}, \quad y(s) = \frac{C(t)}{C_{50}}, \quad z(s) = \frac{M(t)}{C_{50}}.$$

Therefore $\frac{d}{dt} = r \frac{d}{ds}$.

Tumor equation (nondimensionalization)

Starting from (1)

$$\frac{dT}{dt} = r T \left(1 - \frac{T}{K}\right) - \kappa \frac{C}{C + C_{50}} T.$$

Substitute $T = Kx$, $C = C_{50}y$ and divide by rK . Define

$$\alpha := \frac{\kappa}{r}.$$

This yields

$$\boxed{\frac{dx}{ds} = x(1 - x) - \alpha \frac{y}{1 + y} x.}$$

Effector (C) equation (nondimensionalization)

Starting from (2)

$$\frac{dC}{dt} = pC \cdot \frac{T^{h_1}}{T^{h_1} + T_{50,1}^{h_1}} - \ell C - \delta_C C - qC \cdot \frac{T/C}{T/C + s}.$$

Divide by rC_{50} and substitute the scaled variables. Define the dimensionless groups

$$\beta := \frac{p}{r}, \quad \lambda := \frac{\ell}{r}, \quad \delta_c := \frac{\delta_C}{r}, \quad \gamma := \frac{q}{r}.$$

Also set

$$\theta_1 := \frac{T_{50,1}}{K}.$$

Rewrite the exhaustion-like ratio:

$$\frac{T/C}{T/C + s} = \frac{Kx/(C_{50}y)}{Kx/(C_{50}y) + s} = \frac{x}{x + \mu y}, \quad \mu := \frac{sC_{50}}{K}.$$

Therefore the nondimensional effector equation is

$$\boxed{\frac{dy}{ds} = \beta y \cdot \frac{x^{h_1}}{x^{h_1} + \theta_1^{h_1}} - (\lambda + \delta_c) y - \gamma y \cdot \frac{x}{x + \mu y}.$$

Memory (M) equation (nondimensionalization)

Starting from (3)

$$\frac{dM}{dt} = \ell C - bM \frac{T^{h_2}}{T^{h_2} + T_{50,2}^{h_2}} - \delta_M M.$$

Divide by rC_{50} and define

$$b' := \frac{b}{r}, \quad \delta_m := \frac{\delta_M}{r}, \quad \theta_2 := \frac{T_{50,2}}{K}.$$

We obtain

$$\boxed{\frac{dz}{ds} = \lambda y - b' z \cdot \frac{x^{h_2}}{x^{h_2} + \theta_2^{h_2}} - \delta_m z.$$

Final nondimensional system

Collecting all three equations:

$$\boxed{\begin{aligned} \frac{dx}{ds} &= x(1-x) - \alpha \frac{y}{1+y} x, \\ \frac{dy}{ds} &= \beta y \cdot \frac{x^{h_1}}{x^{h_1} + \theta_1^{h_1}} - (\lambda + \delta_c) y - \gamma y \cdot \frac{x}{x + \mu y}, \\ \frac{dz}{ds} &= \lambda y - b' z \cdot \frac{x^{h_2}}{x^{h_2} + \theta_2^{h_2}} - \delta_m z. \end{aligned}}$$

Dimensionless parameter definitions

$$\begin{aligned}\alpha &= \frac{\kappa}{r}, & \beta &= \frac{p}{r}, & \lambda &= \frac{\ell}{r}, \\ \delta_c &= \frac{\delta_C}{r}, & \delta_m &= \frac{\delta_M}{r}, & \gamma &= \frac{q}{r}, \\ b' &= \frac{b}{r}, & \theta_i &= \frac{T_{50,i}}{K} \ (i = 1, 2), & \mu &= \frac{s C_{50}}{K}.\end{aligned}$$

Assumptions / notes

- Time was nondimensionalized with $\tau = 1/r$. Choosing another time scale (e.g. $1/p$) is straightforward.
- Both C and M were scaled by C_{50} for simplicity.
- Hill functions are dimensionless because T was normalized by K .
- The exhaustion-like term was rewritten as $\frac{x}{x + \mu y}$. If the original form differs, change the algebra accordingly.